

BRECHETEAU RESEARCH INSTITUTE

CHRONON-1

Experimental Pre-Registration of the Chronon Field Residual

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Experimental pre-registration of CHRONON-1 (vertical clock residual slope test)

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Abstract — We pre-register a falsifiable test of a dimensionless signed slope ε_Φ after full GR subtraction, adopting the operational convention $d\tau = \Phi^{-1}dt$ (dt : coordinate/UTC time; τ : local proper time). While some corpus notes use the reciprocal $\Phi \leftrightarrow 1/\Phi$, all registered observables depend only on $\Delta \ln \Phi$ (slopes) and are invariant under this reciprocal convention.

Design sensitivity (v1.0; 95% CL): $|\varepsilon_\Phi|_{95} \sim 1 \times 10^{-4} - 1 \times 10^{-3}$ (lab-dependent), for $\Delta h_{\text{rms}} \in [10, 30]$ m, $N \in [20, 60]$, and Allan deviation $\sigma_y(\tau = 3600 \text{ s}) \sim 10^{-18}$ (order-of-magnitude; see Sec. 3).

Primary endpoint: Detection of a non-zero ε_Φ declared at 5σ under the pre-registered invariances and gates (Secs. 4–5).

Secondary/Diagnostic: vertical qubit coherence (T_2 ; secondary) and clock network common-mode (diagnostic).

Principle: all signals are GR-corrected; any remaining signed slope vs X_0 (Sec. 2) that survives the gates is compatible with a non-metric contribution under the registered invariances. Geometry and null cones remain unchanged; detection is strictly differential.

Keywords — *Chronon Field; temporal coherence; differential metrology; optical clocks; GR subtraction; signed slope ε_Φ ; falsifiability; pre-registration.*

I — Registered Scope, Axioms & Identifiability

Registered scope (frozen experimental contract). This pre-registration fixes design, preprocessing, blinding, primary/secondary endpoints, acceptance gates, and the decision rule. All claims are restricted to the primary endpoint (ε_Φ) from vertical clock comparisons.

Axioms (Experimental setup). The registered observables are governed by the operational relationship:

$$d\tau = \Phi^{-1}(x) dt. \quad (1)$$

Units & Dictionary: Φ [1]; $\ln \Phi$ [1]; $\nabla \ln \Phi$ [m⁻¹]; $\Delta \ln \Phi$ [1]; Γ [s⁻¹].

Detection is strictly differential and post-GR subtraction.

Theoretical backdrop (non-gating): The underlying currents $J^\mu = \Phi u^\mu$ and the divergence $\partial_\mu J^\mu = \Gamma(x)$ provide the conservation framework but are not required as priors for the registered endpoint.

Identifiability & invariants. Progressive Validation Gates (G1–G6) are designed to eliminate path-specific artifacts and local clock-bias fields. Any residual cross-site slope Y vs X that survives these gates is compatible with a non-metric contribution. Invariances *kill* local/path artifacts but cannot correct a wrong GR subtraction; therefore GR-model non-uniqueness is treated as a *systematic* uncertainty carried into σ_X via a frozen ensemble spread (Sec. 4). Only differential GR-subtracted observables are admissible.

Gates & Endpoints. Gates (Summary): G1 Witnesses, G2 A/B Inversion, G3 Link Swap, G4 Time-shift/Shuffle, G5 Covariate Veto, G6 Ablation Suite.

Primary (M1). Two-sided *signed* slope ε_Φ in Y vs X . *Decision rule:* Sec. 5.

Secondaries (M2). (i) Qubits: signed β in T_2^{-1} . (ii) Networks: common-mode spectral excess (Diagnostic C0; Sec. 12).

II — Introduction & Hypothesis

Context & Mechanics. This experiment tests for a non-metric residual after full General Relativity (GR) subtraction. We compare vertical clock rates at different heights, subtract the GR prediction based on local geodesy, and test if the residual scales linearly with the potential offset. In a small domain Δh , a first-order linear slope captures the primary effect.

Conventions (single source). Coordinates $x^\mu = (ct, \mathbf{x})$; 4-divergence $\partial_\mu = (\frac{1}{c}\partial_t, \nabla)$; four-velocity $u^\mu = (\gamma c, \gamma \mathbf{v})$. Covariant laws keep c explicit; weak-field estimates restore units via the dictionary g/c^2 .

Hypothesis. We test the null hypothesis H_0 against a specific alternative H_1 :

- H_0 : $\varepsilon_\Phi = 0$. The residual is pure noise or instrumental offset α .
- H_1 : $\varepsilon_\Phi \neq 0$, invariant under site/operator permutations and link swaps, and surviving the GR-model ensemble spread (see Sec. 4).

A non-zero ε_Φ indicates a contribution compatible with a non-metric tempo field $\Phi(x)$.

Parameter mapping to Stage 00. The Stage 00 multiplicative model $y_{\text{meas}} = y_{\text{GR}}(1 + s \cdot \varepsilon) + \alpha$ is mapped to the registered observables $X_0 = g\Delta h/c^2$ and $Y_0 = y_{\text{meas}} - X_0$. This yields the linear mapping:

$$Y_0 = \alpha + \varepsilon_\Phi X_0 + \text{noise}, \quad \text{where } \varepsilon_\Phi = s \cdot \varepsilon. \quad (2)$$

Sign conventions are fixed by upward steps $\Delta h > 0$.

Working hypothesis. After full subtraction of the GR redshift $y_{\text{GR}} = g\Delta h/c^2$, a residual proportional to $g\Delta h/c^2$ may remain. We define the GR-subtracted residual

$$\delta y_{\text{res}} \equiv y_{\text{meas}} - \frac{g\Delta h}{c^2}, \quad X_0 \equiv \frac{g\Delta h}{c^2}, \quad Y_0 \equiv \delta y_{\text{res}}. \quad (3)$$

Definitions & measurables (single source). Let $y \equiv \delta\nu/\nu$ be the fractional frequency difference between two vertically separated clocks (height offset $\Delta h > 0$, upward). Let y_{meas} denote the measured fractional offset before GR subtraction but after all instrumental and transfer-link corrections.

Units & Signs (operational)

Sign. Upward step $\Delta h > 0$. $X_0 \equiv g\Delta h/c^2$, $Y_0 \equiv y_{\text{meas}} - X_0$. A positive ε_Φ means Y_0 increases with Δh (measured offset $>$ GR prediction).

Units. Φ is **dimensionless** (tempo factor). $[\varepsilon_\Phi] = 1$, $[\beta] = \text{s}^{-1}$, $[\xi] = \text{s}^{-1}$, $g/c^2 \simeq 1.09 \times 10^{-16} \text{ m}^{-1}$.

GR intact. Light cones and $T_{\mu\nu}$ unchanged; detection is *strictly differential* (post-GR subtraction).

Weak-field dictionary. $\partial_z \ln \Phi \simeq g/c^2$, $\Delta \ln \Phi \simeq (g/c^2) \Delta h$.

Loss channel.

$$\Gamma(x) = \Gamma_{\text{env}} + \xi \Phi(x) + \beta \Delta \ln \Phi, \quad \Delta \ln \Phi \simeq \frac{g}{c^2} \Delta h \quad (\text{vertical weak-field increment}). \quad (4)$$

β handles gradient-exposure rate (s^{-1}). Relationship: $\Delta \ln \Phi \approx |\nabla \ln \Phi| \Delta h$. *Domain.* This ansatz is used only for (i) co-located vertical stacks and (ii) controlled exposure profiles.

Scope and invariance. $\Phi(x)$ does not modify light cones nor introduce any additional $T_{\mu\nu}$; GR geometry and energy accounting remain intact. All claims are strictly *differential*: only GR-subtracted slopes, spatial increments $\Delta \ln \Phi$ and inter-epoch ratios $\dot{\Phi}/\Phi$ are observable; a constant Φ is operationally indiscernible (pure re-timing; see Sec. 12).

Abbreviations: WLS = Weighted Least Squares; HAC = Heteroskedasticity- and Autocorrelation-Consistent; EIV = Errors-In-Variables; DD = Dynamical Decoupling.

III — Experimental Objectives

Use X_0 and Y_0 as defined once in Sec. 2. Multiplicity handled in Sec. 8.

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Test type	Observable	Domain	Target	Level (Claim)	Detection criterion
Vertical optical clocks	GR-subtracted residual δy_{res}	0–30 m	ε_{Φ}	Primary (C1)	Signed slope of δy_{res} vs Δh , invariant under link swaps and site permutations $A \leftrightarrow B$; witnesses $(\Delta h = 0)$ pass.
Qubits (cQED / ions)	T_2, T_1, T_{ϕ}	Stacked/hoist	$\beta \text{ (s}^{-1}\text{)}$	Secondary (C1)	Monotone variation $T_2^{-1} = T_{\phi}^{-1} + \frac{1}{2T_1} + \beta \Delta \ln \Phi$; stable under DD-toggles; Holm–Bonferroni (Sec. 8).
Clock network	Coherence $\mathcal{C}(f)$	$10^2\text{--}10^5 \text{ m}$	$\delta \Phi / \Phi$	Diagnostic (C0)	Common-mode in the <i>single pre-registered</i> band (Sec. 12); declare “strong” evidence if $\ln B_{10} \geq 5$ and surrogates/link-independence checks pass. Not a standalone detection and never upgrades the primary claim.

Table 1 Pre-registered detection channels and claim levels.

Realistic sensitivity targets. With $\Delta h_{\text{rms}} \in [10, 30]$ m, $N \in [20, 60]$, and Allan deviation $\sigma_y(\tau)$ evaluated at the chosen window/integration time (e.g. $\tau = 3600$ s),

$$\sigma_{\hat{\varepsilon}_\Phi} \approx \frac{\sigma_y(\tau)}{\sqrt{N}} \frac{c^2}{g \Delta h_{\text{rms}}}, \quad \frac{g}{c^2} \simeq 1.09 \times 10^{-16} \text{ m}^{-1}. \quad (5)$$

Scenario	Δh_{rms}	N	$\sigma_y(\tau = 1 \text{ h})$	$ \varepsilon_\Phi _{95\%}(\text{computed})$
S1 (baseline)	10 m	20	3×10^{-18}	1.2×10^{-3}
S2 (good lab)	30 m	40	1×10^{-18}	9.5×10^{-5}
S3 (ambitious)*	30 m	60	2×10^{-19}	1.6×10^{-5}

* Effective via 4–8 h co-integration (documented Allan stability).

95% limits use $1.96 \sigma_{\hat{\varepsilon}_\Phi}$. Values reflect the stated $\sigma_y(\tau)$, N , and Δh_{rms} .

IV — Methodology & Runbook

Use X and Y as defined in Sec. 2. HAC/Andrews and Deming/WTLS details centralized in Sec. 5. Controls/gates in Sec. 6.

Design (frozen). Vertical staircase with randomized ascents/descents; $N \in [20, 60]$ runs; identical cabling per level; fixed duty cycle; co-located neutral witnesses; link permutations (fiber/GNSS/free-space) scheduled; environment setpoints logged.

Preprocessing (frozen). UTC/PTP disciplining; RTK/laser geodesy with solid/ocean tides and pressure-loading; compute $X_0 = g\Delta h/c^2$ (with g measured locally via absolute gravimeter where possible, else a frozen model stack); form $Y_0 = \delta y_{\text{res}} = y_{\text{meas}} - X_0$; window averages (**fixed** $\tau_w = 120$ s); HAC-ready series. *Non-gating sensitivity:* repeat the full pipeline at $\tau_w \in \{60, 300\}$ s and report the maximal CI if the slope or SE shifts by $> 15\%$ (conservative policy).

Noise metric. σ_Y is computed from the Allan deviation $\sigma_y(\tau)$ of the windowed $y(t) = \delta\nu/\nu$ at the chosen averaging time τ (dimensionless; τ in seconds).

QC policy (SNR). Windows with `link_SNR_dB` < 20 are *not automatically* dropped from the primary endpoint. First run the **QC bias check** below. If no bias is detected, low-SNR windows may be excluded and kept for sensitivity checks.

QC bias check (frozen). Within each run, compute the *Spearman* correlation between `link_SNR_dB` and X_0 (or Δh). Define SNR–geometry dependence if $|\rho_s| > 0.20$ and $p < 0.01$ (two-sided). If dependence is detected, do *not* exclude by SNR; retain all windows and let inverse-variance weighting via $\sigma_y(\tau_w)$ handle noise. If dependence is not detected, windows with `link_SNR_dB` < 20 may be excluded but must remain flagged and must be re-injected in a non-gating sensitivity check.

Uncertainties. σ_Y is derived from Allan deviation at the chosen averaging time τ . σ_X is propagated via quadrature:

$$\sigma_X = \sqrt{\sigma_{\Delta h}^2 (g/c^2)^2 + \sigma_g^2 (\Delta h/c^2)^2 + \sigma_{\text{model-spread}}^2}, \quad (6)$$

where the GR-model term is defined conservatively as

$$\sigma_{\text{model-spread}} \equiv \max \left(\text{SD}\{X_0^{(A)}, X_0^{(B)}, X_0^{(C)}\}, \frac{\max(X_0^{(\cdot)}) - \min(X_0^{(\cdot)})}{2} \right), \quad (7)$$

evaluated per window/run and then carried into the fit (treated as systematic; no “averaging down”).

Frozen GR-realizations (must be reproducible). Compute X_0 under three realizations, archived with exact version strings in the replication package:

- **Realization A:** geoid/gravity model = EGM2008 (exact release tag archived); ocean tide model = FES2014b; solid Earth tide = IERS 2010; pressure loading = inverted-barometer with archived meteorology product (exact dataset + version).

- **Realization B:** geoid/gravity model = XGM2019e (exact release tag archived); ocean tide model = GOT4.10c; solid Earth tide = IERS 2010; pressure loading as in A (exact dataset + version).
- **Realization C:** geoid/gravity model = EIGEN-6C4 (exact release tag archived); ocean tide model = FES2014b; solid Earth tide = IERS 2010; pressure loading as in A.

Any deviation requires a signed-off log with pre/post hashes (Sec. 7).

Regression switch. Switch to *Deming/WTLS* if $\sigma_X/X_{\text{rms}} \gtrsim 0.1$ (ratio λ from instrumented uncertainties). *Exception (co-located stacks):* when $\sigma_{\Delta h} \ll \Delta h_{\text{rms}}$, we keep WLS; small tide/loading jitter entering $X_0(t)$ does not trigger Deming as long as $\sigma_X/X_{\text{rms}} < 0.1$.

GR subtraction ensemble (systematic)

To handle model non-uniqueness, compute X_0 under three frozen realizations (A, B, C). Define $\sigma_{\text{model-spread}} = \max(\text{SD}, \frac{\text{range}}{2})$ across $\{A, B, C\}$, and add it in quadrature to σ_X . This term is treated as *systematic* and is not assumed to average down with more windows.

Blinding. Freeze code/configs (hashes). A blinding owner applies a secret affine mask to the analysis variables:

$$Y_0^{\text{blind}} = Y_0 + b, \quad (8)$$

with unknown constant b (and optionally $X_0^{\text{blind}} = sX_0$ with secret $s \in \{-1, +1\}$). All gates and controls except the final fit are executed on blinded data. Witness labels ($\Delta h \approx 0$) remain identifiable for the G1 gate only. Unblinding occurs once all frozen gates pass and the deviation log is empty for the run.

Let $y(t) = \delta\nu/\nu$; $\Delta h > 0$ is upward; $y_{\text{GR}} = g \Delta h/c^2$; $\delta y_{\text{res}} = y_{\text{meas}} - y_{\text{GR}}$.

1. **Calibrate to UTC/PTP.** Phase-lock all clocks/links to a common timebase (UTC traceable, PTP discipline). Record drift of local references.
2. **Measure Δh with tides correction.** Laser/RTK geodesy with solid Earth/ocean tides & pressure-loading corrections. Estimate $\sigma_{\Delta h}$ per window.
3. **Synchronous acquisition (≥ 1 h).** Record $y(t)$ with common timestamps, duty cycle, link type, environment (temperature, vibration proxies).
4. **Subtract GR prediction.** Compute $y_{\text{GR}} = g \Delta h/c^2$ from contemporaneous $\Delta h(t)$. Form $\delta y_{\text{res}}(t) = y_{\text{meas}}(t) - y_{\text{GR}}(t)$.
5. **Aggregate residuals.** Average δy_{res} over windows (60–300 s) to whiten HF noise; propagate uncertainties; pre-whiten if needed for HAC (Sec. 5).
6. **Permuted ascents/descents.** $N = 20$ –60 staircase runs (A/B swaps: top $\leftrightarrow$ bottom), randomized order; identical cabling per level (*see Fig. 1*).
7. **Neutral witnesses ($\Delta h = 0$).** Co-located oscillators and dummy links; require null within errors before proceeding.
8. **Qubit coherence tracking.** Measure $T_2(\Delta \ln \Phi)$ (and T_1, T_ϕ); log pulse sequences; enforce environmental setpoints; DD toggles planned.
9. **Blinding $\rightarrow$ unblinding.** Keep labels/link types blinded until Sec. 6 checks pass; then *unblind* only for the final fit.
10. **Signed-slope analysis.** Primary analysis is **free-intercept WLS** with **HAC (Newey–West; Andrews bandwidth)**; wild bootstrap as non-gating check; switch to **Deming/WTLS** when $\sigma_X/X_{\text{rms}} \gtrsim 0.1$ (λ from instrumented uncertainties). Robust overlay (Huber) reported as sensitivity — *all rules in Sec. 5*.
11. **Inference and report.** Combine runs with meta-aggregation (Sec. 8); emit ledger row with hashes, tests, and verdict (Sec. 6).

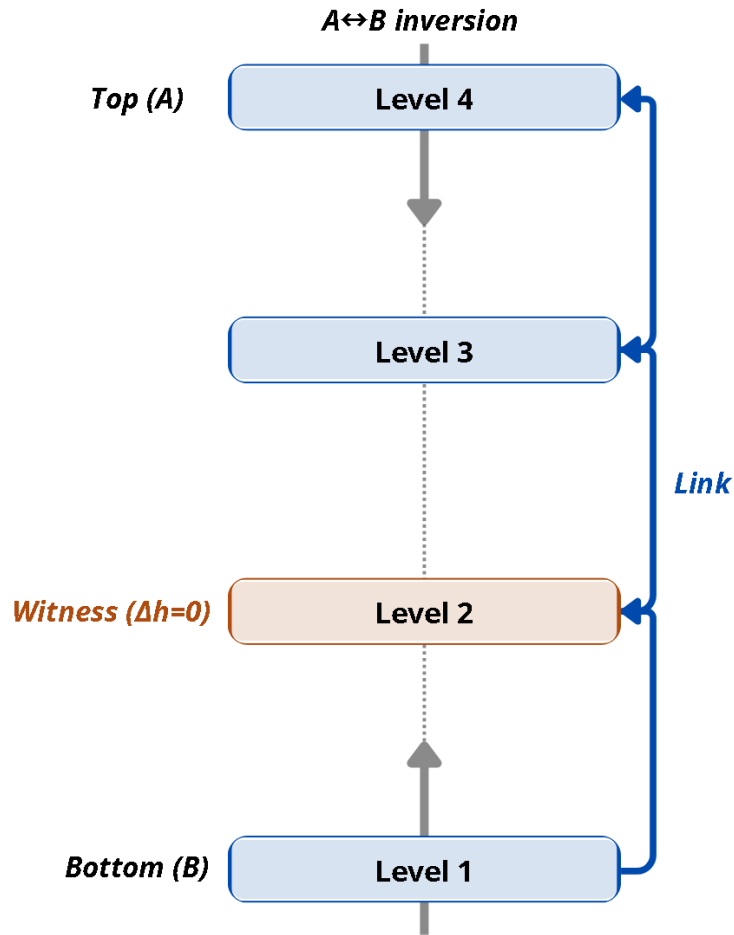


Figure 1 Four-level staircase with witness ($\Delta h = 0$); $A \leftrightarrow B$ inversion and link routing (fiber/GNSS/free-space) shown.

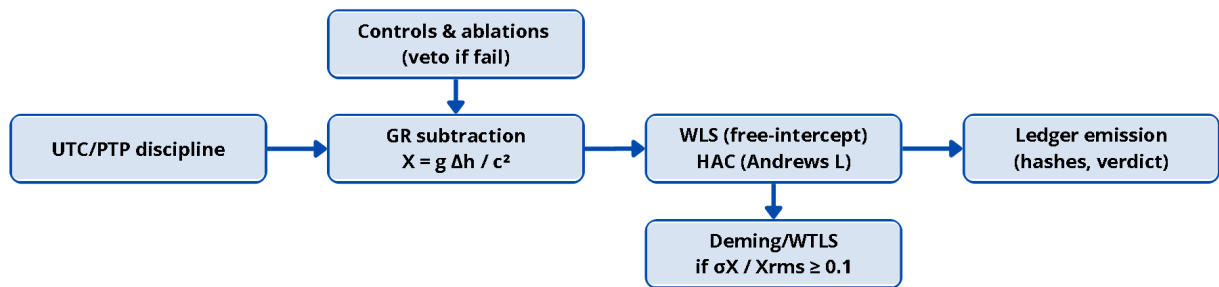


Figure 2 Analysis pipeline: discipline $\rightarrow$ controls/ablations $\rightarrow$ GR subtraction $\rightarrow$ WLS (free-intercept, HAC) $\rightarrow$ Deming/WTLS if $\sigma_X / X_{rms} \geq 0.1$ $\rightarrow$ ledger.

V — Expected Sensitivity & Decision Boundaries

$$Y_i = \alpha + \varepsilon_\Phi X_i + \epsilon_i, \quad (9)$$

with observables defined in Sec. 2. HAC (Newey–West; Bartlett kernel) errors; Andrews plug-in bandwidth L . Report the empirical autocorrelation time τ_c , estimated as the first zero-crossing of the windowed ACF.

HAC bandwidth (Andrews)

Default: $L = \lfloor 4(T/100)^{2/9} \rfloor$, T = number of effective windows.

15% disagreement rule: re-fit with $L/2$ and $2L$. If the resulting 1σ standard errors shift by $>15\%$, the primary SE is broadened to the maximum of the three values. This **conservative policy** is pre-registered to handle bandwidth sensitivity.

Errors-in-variables (Deming/WTLS). Use Deming/WTLS when $\sigma_X/X_{\text{rms}} \gtrsim 0.1$; TLS is $\lambda = 1$.

$$\lambda = \frac{\sigma_Y^2}{\sigma_X^2} \quad \text{fixed from instrumented/propagated uncertainties (geodesy, tides, loading), never from residuals (anti-cir)} \quad (10)$$

Exception. Strictly co-located vertical stacks with $\sigma_{\Delta h} \ll \Delta h_{\text{rms}}$ remain in WLS; Deming/WTLS is only used when $\sigma_X/X_{\text{rms}} \geq 0.1$.

Intercept policy (stability-first). α captures instrumental offsets (e.g. fiber mismatch, residual link bias) and must be (i) constant across permutations/site/operator and (ii) null on witnesses $\Delta h = 0$; weak nonzero α off-witness is tolerated only if (i) holds.

Robust overlay (policy). Report Huber-robust ($k = 1.345$) as *non-gating* sensitivity. *Switch rule:* if $|\hat{\varepsilon}_\Phi|$ shifts $>15\%$ or $\text{SE}_{\text{robust}} > 1.25 \text{SE}_{\text{WLS}}$, adopt the wider CI.

Decision rule (pre-registered, single endpoint)

Verdict: Detection = 5σ two-sided ($|z| \geq 5.0$) *plus* Invariances pass (link/A $\leftrightarrow$ B) *plus* Witness pass *plus* no Veto.

Primary: Two-sided z -test on $\hat{\varepsilon}_\Phi$ (WLS/HAC or Deming/WTLS per rule).

Stability: Intercept α stable and null on witnesses.

If not detected, publish 95% bounds.

Scaling.

$$\sigma_{\hat{\varepsilon}_\Phi} \approx \frac{\sigma_Y}{\sqrt{N}} \frac{c^2}{g \Delta h_{\text{rms}}} \Rightarrow \text{gain via larger } \Delta h, \text{ longer co-integration, and more permutations.} \quad (11)$$

Required integration time (rule-of-thumb).

$$\tau_{\text{req}} \simeq \left(\frac{5 \sigma_y(\tau_0)}{|\varepsilon_\Phi| g \Delta h_{\text{rms}} / c^2} \right)^2 \tau_0, \quad (12)$$

for a target $|\varepsilon_\Phi|$ at given Δh_{rms} and per-window noise $\sigma_y(\tau_0)$.

Power budget. Use Table 1 (Sec. 3) given $\sigma_Y, N, \Delta h_{\text{rms}}$.

VI — Null-Tests & Controls

Progressive Validation Gates (G1–G6)

G1 witness ($\Delta h \approx 0$). Near-zero definition: $X_w \leq 5 \times 10^{-18}$. Gate: $|Y_0| < \max(3\sigma_Y, 1.5 \times 10^{-18})$ (hourly) and slope $0 \pm \sigma$.

G2 A/B inversion. Under A/B inversion (top–bottom swap), the fitted slope $\hat{\epsilon}_\Phi$ is invariant within 1σ when recomputing X_0 and Y_0 under the same sign convention (Sec. 2).

G3 link swap. Invariance ($\pm 1\sigma$) across fiber, GNSS, and free-space paths.

G4 temporal integrity / non-simultaneity null. Pair $y(t)$ with $X_0(t + \Delta T)$ for $|\Delta T| > \tau_c$ (estimated correlation time). Requirement: $\hat{\epsilon}_\Phi = 0 \pm \sigma$. Any persistent slope under time-shifts indicates timing/link artifacts $\Rightarrow$ veto.

G5 covariate veto (frozen covariates). Covariate set fixed to $\{T, P, H, a_{\text{rms}}, \text{rf_intrusion_flag}\}$ at lag 0 only, standardized within run. No additional covariate/interaction/lag allowed. Veto if (i) $\hat{\epsilon}_\Phi \rightarrow 0 \pm \sigma$ and (ii) partial $R^2 \geq 0.70$.

G6 ablation suite. Mandatory pass on all destroyed-label tests (Sec. 12).

1. **G1: Neutral witnesses — hard gate**. Scale-invariant: hourly $|Y_0| < \max(3\sigma_Y, 1.5 \times 10^{-18})$ and slope $0 \pm \sigma$. Any witness failure $\Rightarrow$ pipeline freeze and *neutral/falsify* verdict.
2. **G2–G3: Invariances**. $\hat{\epsilon}_\Phi$ invariant within 1σ under link swap and site/operator permutations.
3. **G4: Temporal integrity / non-simultaneity null**. Estimate the empirical correlation time τ_c (Sec. 5). Construct time-shift nulls by pairing $y(t)$ with $X_0(t + \Delta T)$ for $|\Delta T| > \tau_c$ (random ΔT per run). Requirement: $\hat{\epsilon}_\Phi = 0 \pm \sigma$. Any persistent slope under such time-shifts indicates a timing/link artifact $\Rightarrow$ detection veto.
4. **G5: Covariate veto (nested regression; frozen covariates)**. Run the primary estimator on $Y_0 \sim X_0$ and on $Y_0 \sim X_0 + C$, where C is restricted to the fixed set $\{T, P, H, a_{\text{rms}}, \text{rf_intrusion_flag}\}$ at lag 0 only (standardized within run; no interactions, no lags, no selection). Automatic veto if the covariate both (i) cancels the slope $\hat{\epsilon}_\Phi \rightarrow 0 \pm \sigma$ and (ii) explains $\geq 70\%$ of the variance of Y_0 (partial $R^2 \geq 0.70$).
5. **G6: Ablations**. Height-label permutations, time desynchronization, window mixing $\Rightarrow$ slopes $0 \pm \sigma$ (Sec. 12).
6. **Model-swap control**. Refit using alternative realizations (Sec. 4); check stability within the model-spread envelope.

VII — Data Schema & Ledger

Primary CSV (one row = one measurement window).

```
# Geometry / identification
timestamp.UTC,site_id,site_pair,site_pair_label,pair_orientation
operator_id,run_id,height_m,Delta_h_m

# Measurement (clock/qubit)
y_frac,allan_tau_s,allan_sy_per_h
T2_s,T1_s,Tphi_s

# Environment
temp_C,pressure_hPa,humidity_pct,a_rms_ms2,rf_intrusion_flag

# Transfer link
link_type,link_SNR_dB,firmware_tag

# Physical models / corrections
g_local_mps2,geoid_model,geoid_version,tide_model
X_GR_model_id,X_GR_value,sigma_X_modelspread
```

sagnac_applied, sagnac_value, pressure_load_corr

Uncertainties

sigma_dh_m

Software / reproducibility

duty_cycle, software_env, random_seed, Phi_convention

gate_pass_flags, fail_reason_codes, model_realization_id

Notes (schema)

pair_orientation $\in \{\text{top-minus-bottom, A-minus-B}\}$; site_pair_label is a human-readable alias.
g_local_mps2 = local gravity used; geoid_model/geoid_version, tide_model logged for reproducibility; model_realization_id $\in \{A, B, C\}$.

sagnac_applied $\in \{0, 1\}$ with numeric sagnac_value. For strictly co-located vertical tests, SR/Sagnac terms are negligible but *still logged*; long baselines use full corrections (Sec. 12).

SNR handling (frozen). link_SNR_dB is always logged. Low-SNR windows (< 20 dB) are *flagged* but *not automatically excluded* from the primary endpoint; exclusion is allowed only if the pre-registered QC bias check (Sec. 4) shows no correlation between SNR and X_0 (or Δh).

EIV switch (pre-registered). Use **Deming/WTLS** if $\sigma_X/X_{\text{rms}} \gtrsim 0.1$ with $\lambda = \sigma_Y^2/\sigma_X^2$ fixed from instrumented/propagated uncertainties (geodesy, tides, loading) — never from residuals (anti-circularity). Details in Sec. 5.

Derived analysis table.

timestamp_UTC, site_pair, swap_flag, operator_id, run_id

X_GR, Delta_lnPhi

Y_res

sigma_X, sigma_Y

band_id

with $X_{\text{GR}} = g \Delta h / c^2$ and $\Delta \ln \Phi \simeq (g/c^2) \Delta h$.

Units (schema). Δh [m]; $y = \delta\nu/\nu$ [1]; $X = g\Delta h/c^2$ [1]; T_1, T_2, T_ϕ [s]; temp_C [°C]; $g/c^2 \simeq 1.09 \times 10^{-16} \text{ m}^{-1}$.

Ledger of proof (immutable append-only).

raw_hash, code_hash, config_hash, blind_labels

fit_model, lambda_ratio, huber_k

alpha_stability_pass, controls_pass, ablations_pass

eps_hat, eps_se, eps_pval, decision_primary

beta_hat, beta_se

common_mode_lnB10, band_f1_mHz, band_f2_mHz, lnB_star

n_perm, n_boot

verdict, {confirm|neutral|falsify}, ORCID_signoff

Pre-registered thresholds: $\ln B^* = 5$; $n_{\text{perm}} \geq 4999$; $n_{\text{boot}} \geq 1999$.

Deviation plan & governance (frozen)**Roles.** Blinding owner / Analysis runner / Stats reviewer.**Deviation log.** Timestamp; scope; rationale; impact; pre/post hashes; reviewer sign-off.**Resampling sizes.** Permutation ≥ 4999 , bootstrap ≥ 1999 (report seeds/hashes).**Licenses & deposits.** Code MIT; data CC-BY 4.0; Zenodo deposits with SHA-256 of raw/clean bundles.**Licensing & public deposits****Code:** MIT. **Data:** CC-BY 4.0.Archive full snapshots (code, data, figures, ledger) with a Zenodo DOI for `chrononlabmeta`.Outputs include: `lnB_band_table.csv` (network band evidence) and `ledger_row.json` (human-readable ledger extract).**Conflicts & role split.** No conflicts of interest declared. Roles split where possible (biostatistics vs analysis execution) to reduce analytic bias.Any control failure freezes the analysis; the ledger records a *neutral/falsify* verdict and its cause. All hashes are SHA-256.**VIII — Statistical Framework***Primary estimation follows the free-intercept WLS (Sec. 5) with HAC (Newey–West; Bartlett kernel). Multi-step verification via Deming/WTLS and studentized wild bootstrap.***Meta-aggregation & Multiplicity.** Forest plots and inverse-variance aggregates combine per-site estimates into a global $\hat{\epsilon}_\Phi$. Multiplicity control applies **only** to frequentist secondary endpoints that yield families of p -values (e.g., qubit sub-tests across platforms/conditions). For those, we use the **Holm–Bonferroni** procedure to maintain FWER at $\alpha = 0.05$. The network diagnostic uses a *single pre-registered* coherence band and a fixed Bayes-factor threshold ($\ln B^* = 5$); it is reported as diagnostic evidence and does not upgrade the primary claim.**Stopping rule.** Calendar-based; no interim looks or data-dependent peeking. All analysis parameters (bandwidth, kernels, seeds) are frozen prior to unblinding.**IX — Falsification Criteria****Verdict Classes (Mutually Exclusive).** Any result must be assigned to exactly one class in the Ledger:

- **CONFIRM (Code C1):** Primary $|z| \geq 5.0$ AND all Gates G1–G6 PASS.
- **NEUTRAL (Code N1):** Ambiguous data (low SNR, failure to integrate) or minor Gate failure without conclusive falsification. Also applied to null results ($|z| < 5.0$) where the design sensitivity $\sigma_{\hat{\epsilon}_\Phi}$ is insufficient to reach the pre-registered power-threshold.
- **FALSIFY (Code F1):** Power-qualified falsification. Primary $|z| < 5.0$ surviving all gates, provided that the estimated $\sigma_{\hat{\epsilon}_\Phi}$ is below a pre-registered threshold (e.g., capable of detecting $|\epsilon_\Phi| \geq \epsilon^*$ at 80% power). Also assigned to conclusive Gate failure (G1, G5, G6) indicating an instrumental or model-based artifact.

Freeze-and-publish. Any gate/control failure $\Rightarrow$ pipeline freeze, Ledger entry with verdict code, and release of public bounds and hashes.

X — Figures & Annexes

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XI — Final Summary

Contractual Recap. This preregistration locks the detection of ε_Φ via the model $Y_0 = \alpha + \varepsilon_\Phi X_0 + \epsilon$ (using definitions in Sec. 2). A claim requires: (i) $|z| \geq 5.0$, (ii) Pass on Gates G1–G6, and (iii) Stability across link/site permutations. Secondaries (qubits/networks) are supportive only.

Outcomes.

- *Null/Fail:* Publish 95% $|\varepsilon_\Phi|$ bounds; Ledger Class N1 or F1.
- *Detection:* Release full replication package (Class C1); prioritize Decisive Run.

XII — Documentary Reanalysis (Non-Gating)

OUTSIDE PREREGISTERED ENDPOINT

This section contains documentary reanalyses of existing data. They are non-gating for the primary CHRONON-1 preregistration and cannot be used to claim a primary detection. No documentary reanalysis result can be used to upgrade the preregistered claim level; it only informs prioritization for Stage 01/02.

A — Vertical clocks — meta-analysis

Scope. Re-analyze published vertical optical-clock comparisons (height offsets 0.1–30 m) with reported $\Delta\nu/\nu$, site geometry, link type (fiber/GNSS/free-space), and uncertainty budgets.

Predictor and subtraction. For each datapoint i ,

$$X_{0,i} \equiv \frac{g_i \Delta h_i}{c^2}, \quad Y_{0,i} \equiv \delta y_{\text{res},i} = y_{\text{meas},i} - X_{0,i}, \quad (13)$$

with g_i from local geoid/gravimetry and Δh_i corrected for solid Earth/ocean tides and pressure loading. *Sign conventions.* See Sec. 2 (*Units & Signs*). Renaming $X, Y \rightarrow X_0, Y_0$ for vertical tests.

Estimators (pre-registered). *Primary (free-intercept WLS + HAC).* Fit $Y_{0,i} = \alpha + \varepsilon_\Phi X_{0,i} + \epsilon_i$ with inverse-variance weights; HAC (Newey–West) errors, Andrews plug-in bandwidth $L = \lfloor 4(T/100)^{2/9} \rfloor$; sensitivity at $L/2, 2L$; wild-bootstrap check (Andrews–Mammen) *non-gating*. *Secondary (origin-constrained overlay).*

$$\hat{\varepsilon}_\Phi^{(0)} = \frac{\sum_i w_i X_{0,i} Y_{0,i}}{\sum_i w_i X_{0,i}^2}, \quad \text{Var}(\hat{\varepsilon}_\Phi^{(0)}) = \frac{1}{\sum_i w_i X_{0,i}^2}. \quad (14)$$

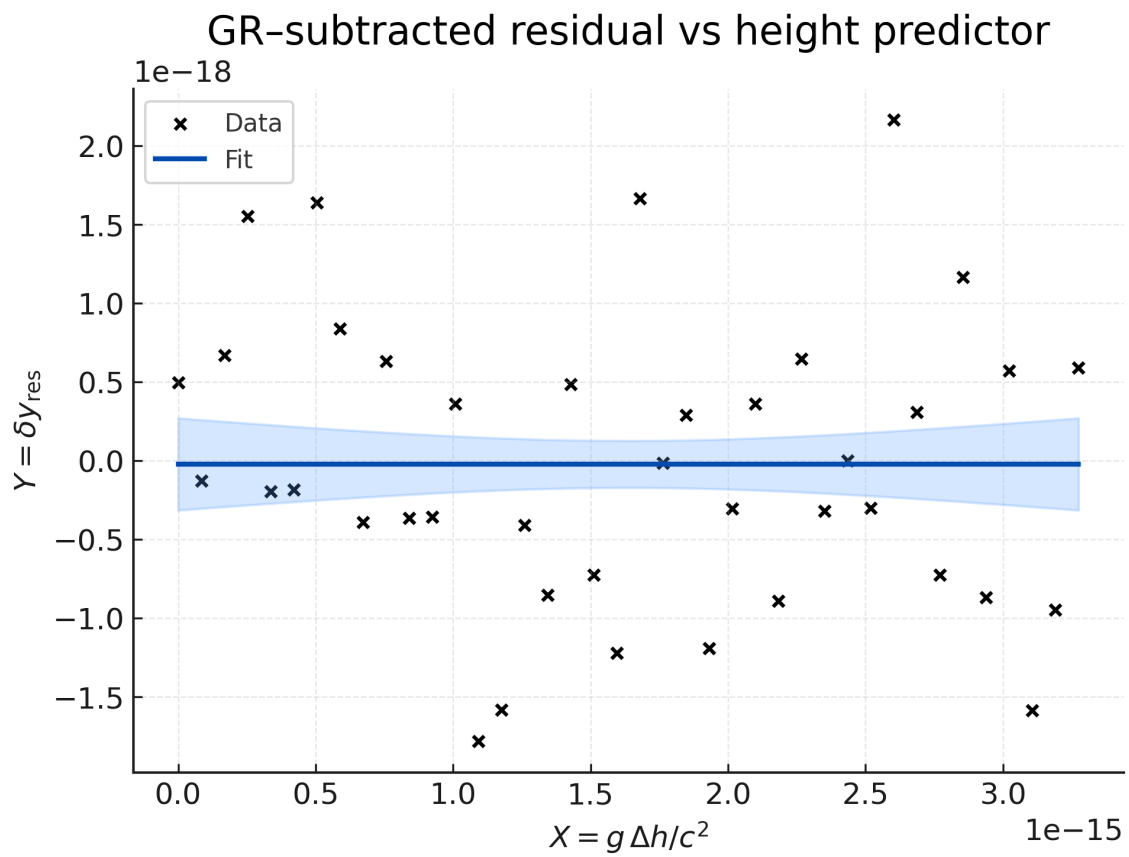


Figure 3 GR-subtracted residual $Y_0 = \delta y_{\text{res}}$ versus $X_0 = g\Delta h/c^2$ fitted by free-intercept WLS with HAC (Andrews bandwidth) and a 95% CI band.

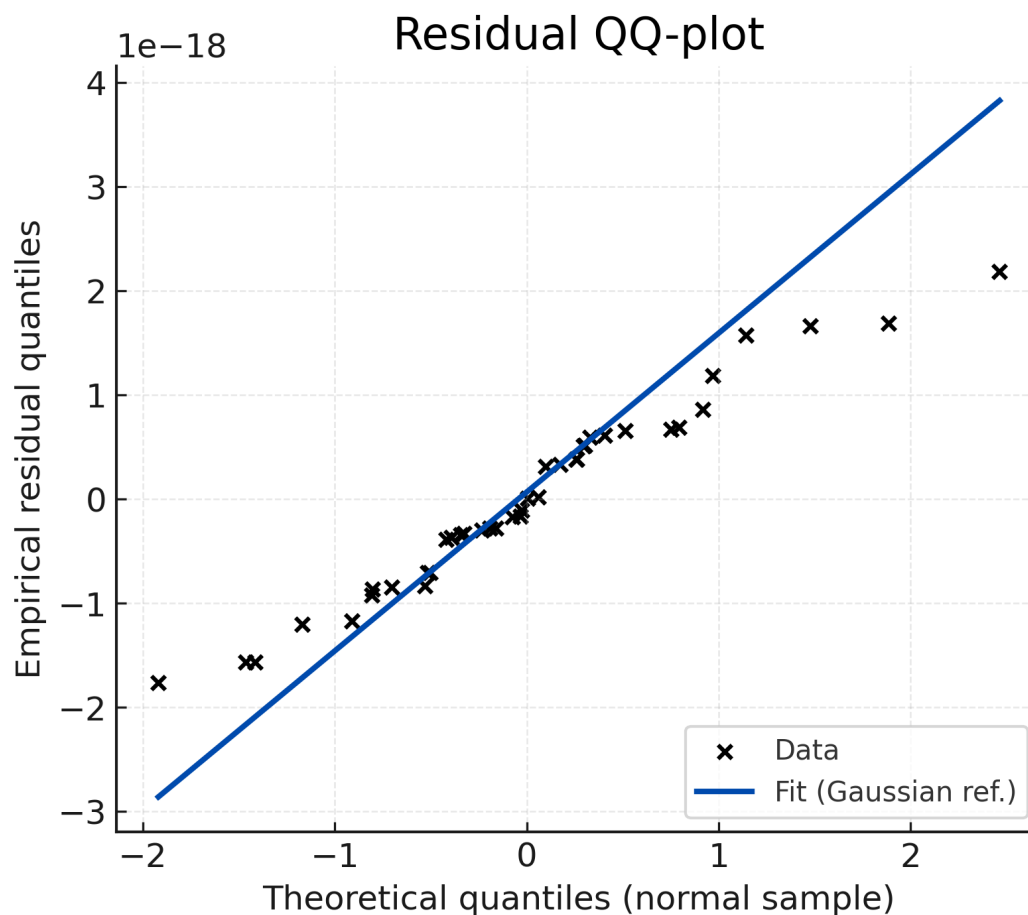


Figure 4 Quantile–quantile plot of WLS residuals (windowed averages) against a Gaussian reference with a dashed identity line.

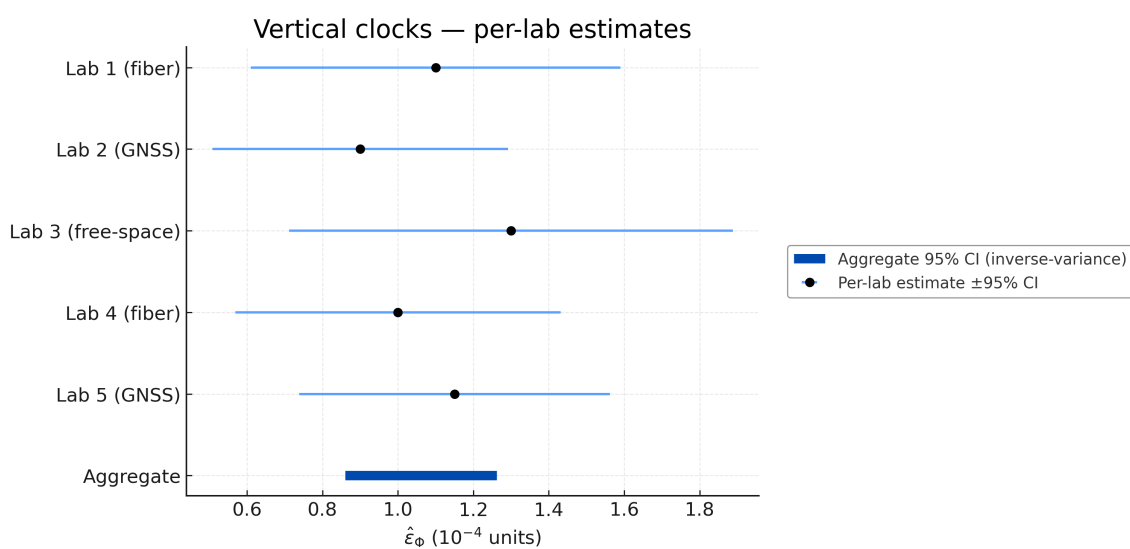


Figure 5 Vertical clocks: per-lab $\hat{\varepsilon}_\Phi$ with 95% CI and inverse-variance aggregate band at the bottom.

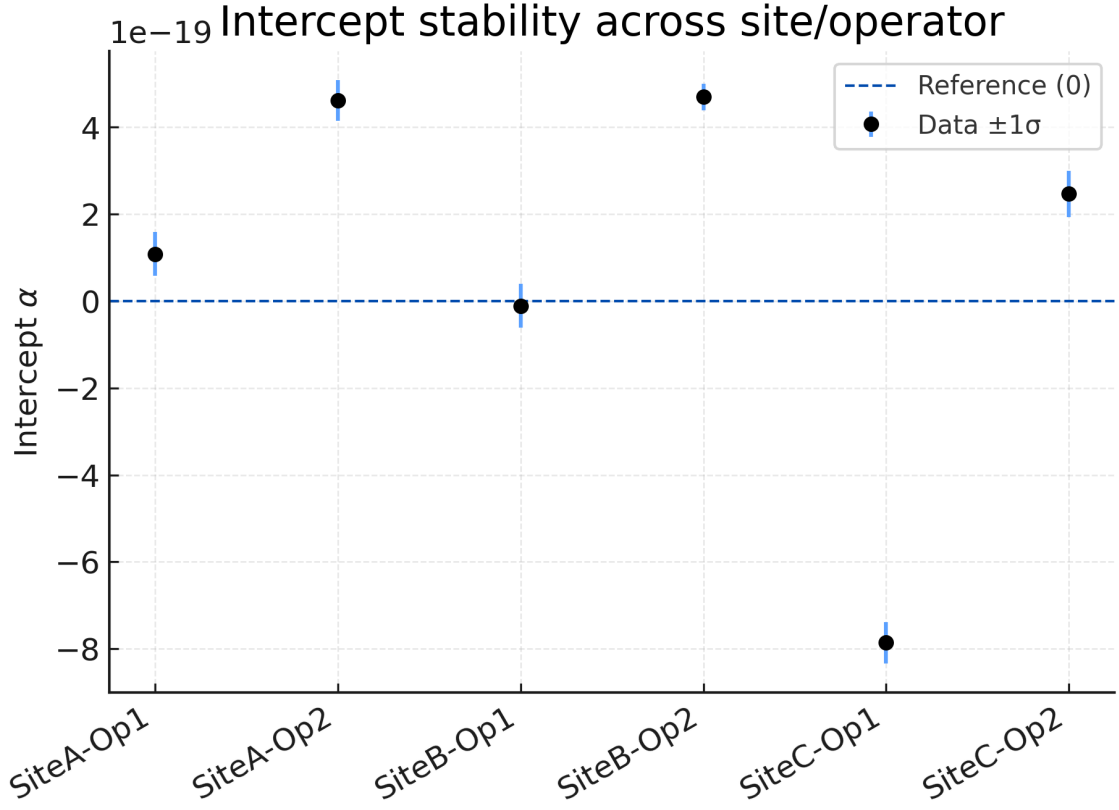


Figure 6 Intercept α across permutations, sites, and operators with a 1σ band and witness panels ($\Delta h = 0$) showing $\alpha \approx 0$.

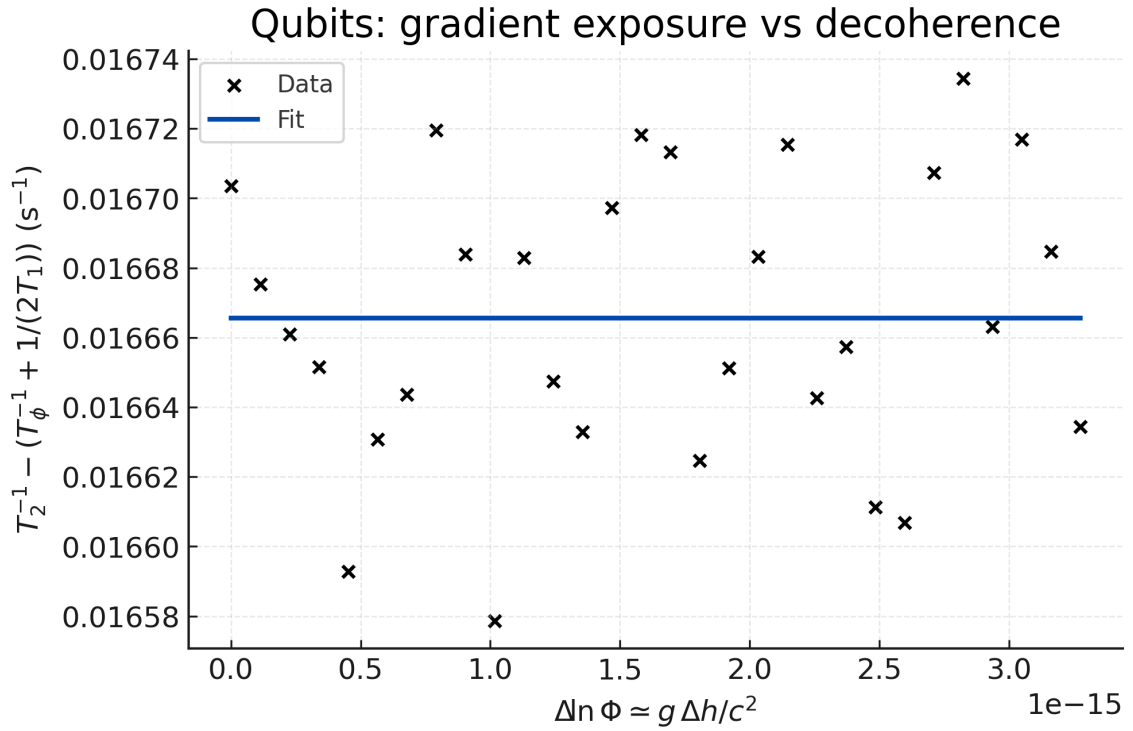


Figure 7 Qubits: T_2^{-1} versus exposure $X_\Phi = \Delta \ln \Phi$ with a free-intercept fit and HAC band, including DD on/off markers.

EIV condition (Deming/WTLS). If $\sigma_{\Delta h}$ is non-negligible ($\sigma_X/X_{\text{rms}} \gtrsim 0.1$), use Deming/WTLS with ratio

$$\lambda = \frac{\sigma_Y^2}{\sigma_X^2} \quad \text{fixed from } \textit{instrumented/propagated} \text{ uncertainties (geodesy, tides, loading)} \text{ — never from residual fits.} \quad (15)$$

For co-located vertical stacks $\sigma_X = (g/c^2) \sigma_{\Delta h}$ with $\sigma_{\Delta h}$ from quadrature of geodesy + solid/ocean tides + pressure-loading models. The *primary verdict follows Deming/WTLS* when this condition holds.

Invariance and null-tests. Report $\hat{\varepsilon}_\Phi$ per link type and site pair; require consistency within 1σ . Permutation tests: (i) height-label shuffles, (ii) non-simultaneous mixes $\Rightarrow$ zero slope within errors. **Witness panels ($\Delta h = 0$) gate:** hourly $|\overline{Y}| < \max(3\sigma_Y, 1.5 \times 10^{-18})$ and slope $0 \pm \sigma$. If any covariate (thermal/vibrational/RF) explains $\geq 70\%$ of the variance of Y and kills the slope, *veto*.

Decision and output. Apply the *single endpoint* of Sec. 5. If no detection, quote

$$|\varepsilon_\Phi| < 1.96 \sigma_{\hat{\varepsilon}_\Phi} \quad (95\% \text{ CL}) \quad (16)$$

per lab and combined (fixed- or random-effects meta-estimate). Deliverables: Y vs X plot (free-intercept primary + origin-constrained overlay), forest plot by lab/link, bounds table (68/95%), HAC vs bootstrap errors, invariance checks.

B — Relativistic geodesy / chronometric leveling

Scope. Reprocess long-baseline clock comparisons used for chronometric leveling (inter-city to continental baselines; mixed relief), using public $\Delta\nu/\nu$, geoid models, and transfer-link metadata.

$$X_{ij}^{\text{GR}} \equiv \frac{\Delta U_{ij}}{c^2} + \frac{\Delta(v^2/2)_{ij}}{c^2} + X_{ij}^{\text{Sagnac}} + X_{ij}^{\text{tides/loads}}, \quad (17)$$

We treat X_{ij}^{GR} as an ensemble of realizations from available geoid/tide/loading models.

Residual and bound.

$$Y_{ij} \equiv \left(\frac{\delta\nu}{\nu} \right)_{ij}^{\text{meas}} - X_{ij}^{\text{GR}} = \left(\frac{\delta\Phi}{\Phi} \right)_{ij} + \epsilon_{ij}. \quad (18)$$

Estimate per-baseline bounds via inverse-variance weighting (or HAC if colored):

$$\left| \frac{\widehat{\delta\Phi}}{\Phi} \right|_{ij} = |\overline{Y_{ij}}|, \quad \text{report 68\%/95\% CL using measured Allan deviation at integration time.} \quad (19)$$

Mapping and binning. Bin baselines by length L and relief Δh :

$$\mathcal{B}_{mn} = \{(i, j) : L \in L_m, \Delta h \in \Delta h_n\}, \quad \text{aggregate } |\widehat{\delta\Phi}/\Phi| \text{ via fixed-effects meta-estimates.} \quad (20)$$

World map (grid 1°) of local upper bounds: assign each site the tightest bound from incident baselines and interpolate conservatively.

Controls & null-tests. (1) *Link invariance:* identical bounds across fiber/GNSS/free-space within 1σ . (2) *Path veto:* remove epochs with anomalous tropospheric/ionospheric indices; bounds must tighten or stay unchanged. (3) *Permutations:* shuffle time tags or pairings $\Rightarrow$ residuals vanish within errors.

Deliverables. Figure: global map of 95% CL bounds on $|\delta\Phi/\Phi|$ with length/relief insets. Tables: per-baseline $|\delta\Phi/\Phi|$ (68/95%), per-bin aggregates, link-type ANOVA p -values.

C — Long-distance common-mode (public time archives)

Scope. Exploit UTC/TAI ensemble series and logs of ultra-stable oscillators (when available) to search for a common-mode fractional-frequency process across distant, *independently linked* sites not sharing the same transfer path.

Data and preprocessing. Construct site-wise $y_k(t) = \delta\nu_k/\nu_k$ on a common grid (1–60 s): detrend steps, remove steering terms, gap-fill conservatively (flagged), pre-whiten (first difference or HP filter) before spectral analysis. **Apply anti-aliasing FIR before any decimation; document the decimation factor and cutoff to avoid diurnal/weekly leakage into [0.1, 10] mHz.**

Spectral/coherence tests. For independent-link sites k, ℓ , estimate one-sided PSDs $S_{y_k}(f)$, $S_{y_\ell}(f)$ and cross-PSD $S_{k\ell}(f)$. Coherence:

$$\mathcal{C}_{k\ell}(f) = \frac{|S_{k\ell}(f)|^2}{S_{y_k}(f) S_{y_\ell}(f)}. \quad (21)$$

Under the null, $\mathcal{C}_{k\ell}(f)$ sits within finite-sample baselines; a bandwidth-stable excess indicates a common mode. Pre-registered band: $[f_1, f_2] = [0.1, 10]$ mHz, excluding diurnal/weekly lines; $\pm 30\%$ band stability required.

Bayesian common-mode model.

$$y_k(t) = a_k s_c(t) + n_k(t), \quad (22)$$

with s_c zero-mean common process (PSD $S_c(f)$), $a_k \approx 1$, n_k independent link/site noises. Compare $H_1 : S_c > 0$ vs $H_0 : S_c = 0$ in frequency space; report aggregated evidence over $[f_1, f_2]$ with the **pre-registered threshold**

$$\boxed{\ln B^* = 5} \quad (\text{“strong” level}). \quad (23)$$

Chronon translation and bound. Spatially coherent fractional-frequency implies a coherent $\delta\Phi/\Phi$. If $\ln B_{10} < \ln B^*$,

$$\left\langle \left| \frac{\delta\Phi}{\Phi} \right|^2 \right\rangle_{[f_1, f_2]} = \int_{f_1}^{f_2} S_c(f) df < \sigma_c^2 \text{ (95\% CL)}, \quad (24)$$

quote $|\delta\Phi/\Phi|_{[f_1, f_2]} < \sigma_c$ at regional/continental scales.

Controls and nulls. (1) *Surrogates*: circular time-shifts and phase randomization return coherence to baseline.

(2) *Link independence*: repeat on alternative links; any persistent peak must survive path changes.

(3) *Band stability*: peaks persist under moderate $[f_1, f_2]$ and windowing variations.

Deliverables. Figure: $\mathcal{C}_{k\ell}(f)$ with null bands and vetted region; inset $|S_{k\ell}(f)|$.

Table: $\ln B_{10}$ per pair and combined; 95% CL bounds on $|\delta\Phi/\Phi|$ (band-limited).

D — Qubits — documentary reanalysis

Scope. Re-examine published coherence datasets T_2 (with T_1 , T_ϕ controls) from circuit QED and trapped-ion platforms that include: (i) stacked levels with known Δh ; (ii) elevator/hoist/tower moves; (iii) parabolic-flight or centrifuge runs with controlled $g_{\text{eff}}(t)$ profiles.

Chronon model (Option A — dimensionless exposure).

$$\Gamma = \Gamma_{\text{env}} + \xi \Phi + \beta \Delta \ln \Phi, \quad \Delta \ln \Phi \equiv \int_C (\partial_n \ln \Phi) dl. \quad (25)$$

Static vertical stacks: $\Delta \ln \Phi \simeq (g/c^2) \Delta h$. Profiled accelerations: $\Delta \ln \Phi_{\text{eff}} \simeq \int (g_{\text{eff}}(t)/c^2) v_n(t) dt$.

Predictors and regressions. For each controlled segment k ,

$$X_{\Phi,k} \equiv \Delta \ln \Phi_k, \quad Y_{\Phi,k} \equiv T_{2,k}^{-1} - \left(T_{\phi,k}^{-1} + \frac{1}{2T_{1,k}} \right). \quad (26)$$

Primary fit (sensitivity): free-intercept WLS with HAC (Newey–West; Andrews bandwidth, policy Sec. 5); studentized wild bootstrap (Andrews–Mammen) as non-gating check.

Origin-constrained overlay (diagnostic):

$$\hat{\beta}^{(0)} = \frac{\sum_k w_k X_{\Phi,k} Y_{\Phi,k}}{\sum_k w_k X_{\Phi,k}^2}, \quad \text{Var}(\hat{\beta}^{(0)}) = \frac{1}{\sum_k w_k X_{\Phi,k}^2}, \quad w_k = 1/\sigma_{Y,k}^2. \quad (27)$$

Separating ξ from β . Across epochs/platforms with identical X_k but offset shifts, absorb platform-wise shifts into $\xi \Phi$ (intercepts) while keeping cross-condition slope for β . Validate with DD toggles: ξ -like offsets respond to DD; a true $\beta \Delta \ln \Phi$ term is DD-robust.

Controls and nulls. Hold cryostat T , magnetic map, flux-bias, phonon environment fixed; veto segments with T_1 or T_ϕ excursions $> 1\sigma$. Negative controls: exposure shuffles; non-simultaneous mixes; synthetic $1/f^\alpha$ injections $\Rightarrow$ zero slope within errors.

Inference and output. Report $\hat{\beta} \pm 1\sigma$; apply the multiplicity policy (Holm–Bonferroni; Sec. 8). If null, publish $|\beta| < 1.96 \sigma_{\hat{\beta}}$ (95%). Deliverables: T_2^{-1} vs X figure (free-intercept + origin-constrained overlay); DD on/off comparison; table of $\hat{\beta}$, controls, and platform meta-estimate.

E — Cosmology — Separate Speculative Series

Scope (Publish Separately; Separate DOI). This subsection is part of a separate speculative series and is not part of the CHRONON-1 preregistered claim.

Priors (recall). Flat: $\Omega_m \in [0.1, 0.5]$, $H_0 \in [50, 90] \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\varepsilon_\Phi^{\cos} \in [-0.1, 0.1]$.

Scope. Implement the rhythmic mapping

$$H(z) \equiv -\frac{\dot{\Phi}}{\Phi}, \quad a(t) \propto \Phi^{-1}(t), \quad (28)$$

and test minimal deformations of the expansion history against public $H(z)$, anisotropic BAO, strong-lens time delays, and SN *shape-only* (no absolute magnitude).

Datasets and predictors. Use $\{E(z) = H(z)/H_0\}$ (cosmic chronometers), anisotropic BAO $\{D_A/r_s, c/(Hr_s)\}$, Alcock–Paczyński

$$F_{\text{AP}}(z) = (1+z) \frac{D_A(z)H(z)}{c}, \quad (29)$$

strong-lens $D_{\Delta t}H_0/c$, and SN Hubble diagram with free M_B (shape-only).

Model families (pre-registered).

- $\mathcal{M}_0$: flat Λ CDM cast into the same dimensionless observables.
- $\mathcal{M}_\Phi$: minimal rhythmic deformation

$$E_\Phi(z) = E_{\Lambda\text{CDM}}(z) [1 + \varepsilon_\Phi^{\cos} f(z)], \quad f(z) \in \{\ln(1+z), z/(1+z)\}, \quad f(0) = 0. \quad (30)$$

Fit. Joint likelihood with published covariances (block-diagonal across probe families); SN block marginalized over (M_B, α, β) . HAC/Andrews and wild-bootstrap policies as in Sec. 5 for any time-series components.

Decision and reconstruction. Preference via $\Delta\chi^2 \equiv \chi^2(\mathcal{M}_0) - \chi^2(\mathcal{M}_\Phi)$.

$$\boxed{\Delta\chi^2 > 6 \text{ and } \text{sign}(\hat{\varepsilon}_\Phi^{\text{cos}}) \text{ stable under probe-split cross-validation.}} \quad (31)$$

From the posterior,

$$\frac{\dot{\Phi}}{\Phi}(z) = -H_0 E_\Phi(z), \quad \ln \frac{\Phi(z)}{\Phi_0} = - \int_0^z \frac{E_\Phi(z')}{1+z'} dz'. \quad (32)$$

Degeneracies (checks). (1) H_0 – M_B neutralized by SN shape-only; cross-check with BAO/AP and $D_{\Delta t} H_0/c$.
(2) Ω_m – $\varepsilon_\Phi^{\text{cos}}$ broken by combining D_A/r_s with $c/(Hr_s)$ and adding F_{AP} .

Deliverables. Figure: posterior $E_\Phi(z)$ vs $E_{\Lambda\text{CDM}}(z)$ with 1σ bands; inset $\ln[\Phi(z)/\Phi_0]$. Table: $\hat{\varepsilon}_\Phi^{\text{cos}} \pm 1\sigma$, $\Delta\chi^2$, probe-split stability, and 95% bounds.

F — Precision spectroscopy — retrodiction

Claim (minimal universality). If $\Phi(t)$ is spatially uniform over an epoch $t_0 \pm \Delta t$ and acts universally, then laboratory spectra co-scale with Φ and *dimensionless* ratios are invariant:

$$\nu_a(t) = \Phi(t) \nu_a^{(0)}, \quad R_{ab}(t) \equiv \frac{\nu_a(t)}{\nu_b(t)} = \frac{\nu_a^{(0)}}{\nu_b^{(0)}} = \text{const.} \quad (33)$$

Hence, intra-epoch spectroscopy cannot detect a constant Φ (pure re-timing; see also Sec. 12 for gradients-only observables).

Inter-epoch test (anchored to a standard). With a stable reference S (e.g., TAI/Cs ensemble),

$$y_a(t) \equiv \frac{\nu_a(t)}{\nu_S(t)} - 1 \simeq -\frac{\dot{\Phi}}{\Phi} t + n_a(t) - n_S(t), \quad (34)$$

so a secular drift in y_a bounds $|\dot{\Phi}/\Phi|$ over multi-year baselines.

Estimators and bounds. Fit $y_a(t) = \alpha_a + \beta_a t + \epsilon(t)$ by free-intercept WLS with HAC errors (Andrews bandwidth; policy Sec. 5) and wild bootstrap (Andrews–Mammen; non-gating). Quote

$$|\dot{\Phi}/\Phi| < 1.96 \sigma_{\hat{\beta}_a} \quad (95\% \text{ CL}) \quad (35)$$

per species; combine by inverse-variance meta-estimate. Universality check: optical–optical ratios $R_{ab}(t)$ must satisfy $\frac{d}{dt} \ln R_{ab} = 0$ within errors.

Dataset requirements & deliverables. Long baselines ($\gtrsim$ year), documented steering of S , environmental logs; include Al^+ , Sr , Yb , Hg . Deliverables: $y_a(t)$ with linear/piecewise fits; residual histogram and HAC window; table of per-species $\hat{\beta}_a \pm 1\sigma$ and the combined bound on $|\dot{\Phi}/\Phi|$; optical–optical $\hat{R}_{ab}$ checks.

G — Equivalence theorem (partial no-go)

Statement.

- (i) **Re-timing equivalence.** If $\Phi(x) = \Phi_0$ is constant on a spacetime domain $\mathcal{D}$, all admissible observables in $\mathcal{D}$ are invariant under $t' = \Phi_0 t$; thus Φ_0 is operationally indiscernible (pure re-timing).
- (ii) **Gradient-only observables.** Detectable signatures require spatial increments $\Delta \ln \Phi \neq 0$ (co-located differential tests) or inter-epoch variation $\dot{\Phi}/\Phi \neq 0$ (multi-epoch drifts). Otherwise, null by (i).

Proof sketch. The master laws

$$d\tau = \Phi^{-1} dt, \quad J^\mu = \Phi u^\mu, \quad \partial_\mu J^\mu = \Gamma \quad (36)$$

are covariant under $t \mapsto t' = \Phi_0 t$ when $\Phi = \Phi_0$. Any phase integral $\varphi = \int \omega dt$ becomes $\varphi = \int (\omega/\Phi_0) dt'$: uniform rescaling of all rates. Dimensionless ratios are unchanged; GR observables are unmodified since Φ neither alters the metric nor adds stress–energy.

Weak-field dictionary (operational).

$$\partial_z \ln \Phi \simeq \frac{g}{c^2} \quad (\text{m}^{-1}), \quad \Delta \ln \Phi \simeq \frac{g}{c^2} \Delta h \quad (\text{dimensionless}), \quad \frac{g}{c^2} \simeq 1.09 \times 10^{-16} \text{ m}^{-1}. \quad (37)$$

Consequence. All CHRONON-1 claims must be **differential** (signed slopes vs $X_0 = g\Delta h/c^2$; $\Delta \ln \Phi$ exposures; inter-epoch drifts) and survive invariances/controls; otherwise re-timing artifacts. Sign conventions: see Sec. 2 (*Units & Signs*).

H — Pre-registered simulations (false-positive calibration)

Purpose. Quantify the false-positive rate and statistical power *before* any new data, using open simulations with re-injected real noise.

Signal+noise model. Generate paired series on a grid $\Delta t \in [1, 60]$ s:

$$X_0(t) = \frac{g \Delta h(t)}{c^2}, \quad y_{\text{meas}}(t) = X_0(t) + \varepsilon_\Phi X_0(t) + n(t) + s(t), \quad Y_0(t) = y_{\text{meas}}(t) - X_0(t) = \varepsilon_\Phi X_0(t) + n(t) + s(t), \quad (38)$$

where $n(t)$ mixes white/flicker/random-walk FM (weights tuned to target $\sigma_y(\tau)$); $s(t)$ are optional small systematics for stress tests.

Qubit branch: $X_\Phi(t) = \Delta \ln \Phi(t)$, $Y_\Phi(t) = T_2^{-1} - (T_\phi^{-1} + \frac{1}{2T_1})$ with parameter β .

Parameter sweeps (pre-registered).

$$\varepsilon_\Phi \in [10^{-4}, 10^{-3}], \quad \beta \in [10^{-4}, 10^{-2}] \text{ s}^{-1}, \quad N_{\text{runs}} \in \{10, 20, 40\}, \quad \Delta h_{\text{rms}} \in \{5, 10, 20, 30\} \text{ m}. \quad (39)$$

Analysis pipeline

Compute $Y_{\text{res}} = y_{\text{meas}} - X$.

Primary: free-intercept WLS with HAC (Newey–West; Andrews plug-in bandwidth $L = \lfloor 4(T/100)^{2/9} \rfloor$).

Sensitivity: re-fit with $L/2$ and $2L$; widen CI if $> 15\%$ spread.

EIV switch: Deming/WTLS if $\sigma_X/X_{\text{rms}} \gtrsim 0.1$ with $\lambda = \sigma_Y^2/\sigma_X^2$ from instrumented uncertainties (never residuals).

Wild bootstrap (Andrews–Mammen) $N_{\text{boot}} \geq 1999$ (non-gating; widen CI if disagreement $> 15\%$).

Vertical co-located uncertainty.

$$\sigma_X = \frac{g}{c^2} \sigma_{\Delta h}, \quad \sigma_{\Delta h} = \sqrt{\sigma_{\text{geod}}^2 + \sigma_{\text{tides}}^2 + \sigma_{\text{load}}^2}. \quad (40)$$

Controls embedded in sims. Include link swaps, A/B inversions, witness panels ($\Delta h = 0$), and covariate injections (thermal, vibrational, RF) with known coupling.

Outputs & pass criteria. ROC curves (TPR vs FPR) for the 5σ rule; null p -histograms (uniform); QQ-plots; operating points vs $\Delta h_{\text{rms}}, N$.

Pass if (all)

1. Empirical FPR at $5\sigma \leq 3 \times 10^{-7}$.
2. Power ≥ 0.8 for $\varepsilon_\Phi \geq 10^{-3} \times \frac{30\text{m}}{\Delta h_{\text{rms}}}$ with $N \geq 20$.
3. Witness gate ($\Delta h = 0$): hourly $|\overline{Y}| < \max(3\sigma_Y, 1.5 \times 10^{-18})$ and slope $0 \pm \sigma$.
4. Covariate veto: any covariate explaining $\geq 70\%$ of Y and killing the slope triggers a *veto* flag in the Ledger.

I — Ablations & negative controls

Purpose. Demonstrate that the pipeline does not manufacture spurious slopes/coherences; any genuine signal must vanish under label destruction or non-simultaneity.

Ablation suite (pre-registered).

1. **Height-label permutation.** Randomly permute Δh within runs $\Rightarrow \hat{\varepsilon}_\Phi^{\text{perm}} = 0 \pm \sigma$; permutation p -values uniform.
2. **Time desynchronization.** Pair $y(t)$ with $X(t + \Delta T)$ for random ΔT beyond correlation time $\Rightarrow$ slope $0 \pm \sigma$.
3. **Window mixing.** Mix non-overlapping segments across days/runs (same Δh histogram) $\Rightarrow$ slope vanishes; HAC errors match null sims.
4. **Sign inversion.** Flip $\Delta h \rightarrow -\Delta h$ without moving hardware $\Rightarrow$ fitted slope flips sign (diagnoses static systematics).
5. **Covariate regression-out.** Regress y on environmental logs (temp., vibration, RF). **Automatic veto:** a covariate explaining $\geq 70\%$ of Y and killing the slope triggers a veto.
6. **Synthetic $1/f^\alpha$ injections.** Ensure colored noise is not converted into signed slope; $|\hat{\varepsilon}_\Phi| < 1\sigma$.
7. **Negative-height control.** Two co-located heads ($\Delta h = 0$) with cabling/link variations: hourly $|\overline{\delta y_{\text{res}}}| < \max(3\sigma_Y, 1.5 \times 10^{-18})$; slope $0 \pm \sigma$.

Reporting. Histogram of t -statistics (KS vs $\mathcal{N}(0,1)$); permutation p -histogram; HAC bandwidth sensitivity (Andrews $L, L/2, 2L$); robust overlay (Huber $k = 1.345$). Any failure is recorded in the Ledger (Sec. 7); discrepancies $> 15\%$ widen the reported CI (policy Sec. 5).

J — Canonical Bounds 2025 (short, publishable)

Format (one page per axis). Each page reports: *Equation/Model* (testable law), *Dataset* (public sources), *Method* (analysis sketch), *Numerical bound* (68/95%), *Controls* (key nulls), *Reproducibility* (hashes/links).

J.1 — Vertical clocks (co-located, GR-subtracted).

$$\text{Model: } Y = \alpha + \varepsilon_\Phi X + \epsilon, \quad X = \frac{g \Delta h}{c^2}, \quad Y = \delta y_{\text{res}}. \quad \Rightarrow \quad |\varepsilon_\Phi| < \{\varepsilon_{\Phi,68}, \varepsilon_{\Phi,95}\}. \quad (41)$$

J.2 — Clock networks (long baselines / chronometric leveling).

$$Y_{ij} = (\delta\nu/\nu)_{ij}^{\text{meas}} - X_{ij}^{\text{GR}} \quad \Rightarrow \quad \left| \frac{\delta\Phi}{\Phi} \right|_{ij} \lesssim |\overline{Y}_{ij}| \quad (68/95\%). \quad (42)$$

J.3 — Qubits (documentary reanalysis).

$$T_2^{-1} = T_\phi^{-1} + \frac{1}{2T_1} + \beta \Delta \ln \Phi; \quad |\beta| < \{\beta_{68}, \beta_{95}\}. \quad (43)$$

J.4 — Cosmology (dimensionless selection).

$$E_\Phi(z) = E_{\Lambda\text{CDM}}(z) [1 + \varepsilon_\Phi^{\text{cos}} f(z)], \quad f(0) = 0; \quad |\varepsilon_\Phi^{\text{cos}}| < \{\varepsilon_{\Phi,\text{cos},68}, \varepsilon_{\Phi,\text{cos},95}\}. \quad (44)$$

Canonical table (publication-ready).

Axis	Parameter	68%	95%	Notes / Controls
Vertical clocks	$ \varepsilon_\Phi $	$\varepsilon_{\Phi,68}$	$\varepsilon_{\Phi,95}$	link/A $\leftrightarrow$ B/dummy pass
Clock networks	$ \delta\Phi/\Phi $	$\Delta_{\Phi,68}$	$\Delta_{\Phi,95}$	path veto, permutations
Qubits	$ \beta $	β_{68}	β_{95}	DD-robust, env. veto
Cosmology	$ \varepsilon_\Phi^{\cos} $	$\varepsilon_{\Phi,\cos,68}$	$\varepsilon_{\Phi,\cos,95}$	probe-split stable

K — Execution order (fast path)

Milestones (calendar from day $J = 0$).

1. **J+2 — Repo bootstrap.** Create **chronon-lab-meta** (public): simulators, CSV schemas (Sec. 7), notebook for GR subtraction $\Rightarrow \varepsilon_\Phi$; CI to rebuild Sec. 12 figures from processed CSV.
2. **J+5 — First vertical-clock bounds.** Reprocess 2–3 published stair runs; produce Y vs X plots and WLS+HAC slope table; run link/A $\leftrightarrow$ B/dummy invariances (Secs. 4–6). Output: preliminary $|\varepsilon_\Phi|$ 68/95% per lab.
3. **J+10 — Theory notes.** (i) *No-go (constant Φ)* two-page memo (Sec. 12); (ii) cosmology pre-fit (Sec. 12) with SN shape-only + BAO/AP + $D_{\Delta t}H_0/c$. Output: $|\varepsilon_\Phi^{\cos}|$ bound and $\Delta\chi^2$.
4. **J+14 — Short preprint.** “*Bounds without new experiments*” (5–8 pages): methods (Secs. 2, 7, 8), results (A–F), controls (Secs. 6, 12), canonical table (Sec. 12), code/data hashes (Sec. 7).

Pass/fail gates (each milestone). Proceed only if: (a) reproducibility hashes match; (b) invariance checks pass; (c) ablations are null (Secs. 6, 12). Otherwise, stop and issue a falsification or revise pipeline.

L — Interpretation grid (outcomes)

If all tests are null (within errors). No signed slope or coherence co-variation survives controls:

$$\hat{\varepsilon}_\Phi = 0 \pm \sigma, \quad |\beta| < \beta_{95}, \quad \left| \frac{\delta\Phi}{\Phi} \right| < \Delta_{\Phi,95}. \quad (45)$$

Outcome: the Chronon framework *survives by tightening* — you chart where it cannot act (tested domain), upgrade canonical bounds (Sec. 12), and calibrate CHRONON-1 power (Sec. 12).

If there is a weak, consistent preference. Any axis (vertical clocks, networks, qubits, cosmology) shows stable evidence:

$$\text{either } \Delta\chi^2 > 6 \text{ (model selection) or } \ln B_{10} > \ln B^* \text{ (common mode)}. \quad (46)$$

Outcome: *prioritize* a concrete hypothesis (signed ε_Φ or β) and a configuration (height range, link type, platform) for the next decisive run (Sec. 12).

If signals disagree across axes. Cross-axis inconsistency (e.g., clock slope but no network mode, or T_2 effect without vertical slope) is treated as *systematics* until a common signed parameter survives Secs. 4–6–12. Outcome: *falsified for the claimed domain* or sent back to ablations.

If a detection survives all controls. A 5σ signed slope/invariance-persistent result or a DD-robust T_2 co-variation is *potential evidence* for a non-metric coherence field $\Phi(x)$ under the minimal dictionary. Outcome: release the replication package (ledger, code, raw CSV) $\rightarrow$ public CHRONON-1; summarize in Sec. 11.

M — Replication Package Skeleton

Purpose. This section frames the templates and scripts as a replication package skeleton, providing a roadmap for future work rather than presenting scientific claims.

Bundle structure (reproducible layout).

```

chronon-lab-meta/
|  |-- data/
|  |  |-- raw/                      # published CSVs / digitized points
|  |  |-- processed/                # Sec. VII schema-conformant tables
|  |  |  |-- lnB_band_table.csv      # network band evidence (pre-registered band)
|  |  |  '-- expected_residual_curves.csv
|  |  '-- templates/
|  |     '-- primary_schema.csv      # header from Sec. VII (Sec.~\ref{sec:schema})
|-- img/
|  |-- staircase_schematic.pdf
|  |-- staircase_schematic.png
|  |-- metrology_pipeline.pdf
|  |-- metrology_pipeline.png
|  |-- expected_residual_vs_height.pdf
|  |-- expected_residual_vs_height.png
|  |-- expected_T2_vs_height.pdf
|  |-- expected_T2_vs_height.png
|  |-- residual_qqplot.pdf
|  |-- residual_qqplot.png
|  |-- forest_vertical_clocks.pdf
|  |-- forest_vertical_clocks.png
|  |-- intercept_stability.pdf
|  |-- intercept_stability.png
|  |-- map_dPhi_over_Phi_bounds.pdf
|  '-- map_dPhi_over_Phi_bounds.png
|-- notebooks/
|  |-- 01_gr_subtraction_slope_fit.ipynb
|  |-- 02_clock_residual_simulator.ipynb
|  |-- 03_common_mode_bayes.ipynb
|  |-- 04_qubit_T2_gradient_fit.ipynb
|  |-- 05_cosmo_Ephi_fit.ipynb
|  '-- 06_spectroscopy_drift_bounds.ipynb
|-- scripts/
|  |-- 00_ledger_packager.py         # hashes, ledger rows, CI hooks
|  |-- make_fig_staircase.py         # staircase schematic (Deltah)
|  '-- make_pipeline_diagram.py      # metrology pipeline 1e-16-1e-18
|-- configs/
|  |-- analysis.yaml                # prereg params (Secs.~\ref{sec:decisions}--\ref{sec:stats})
|  '-- noise_palette.yaml            # synth weights for sims (Sec.~\ref{sec:sims})
|-- memos/
|  '-- no_go_constant_phi.tex        # 2p formal note (Sec.~\ref{sec:equivalence})
|-- legal/
|  |-- LICENSE_CODE.txt              # MIT
|  '-- LICENSE_DATA.txt              # CC-BY 4.0
'-- ledger/
   |-- runs_ledger.csv               # append-only (Sec.~\ref{sec:schema})
   '-- ledger_row.json               # human-readable extract (per run)

```

Figures & diagrams (auto-built). Staircase (Δh , A/B, paths) $\rightarrow$ staircase_schematic.\{pdf,png\}.

Pipeline (UTC/PTP $\rightarrow$ GR $\rightarrow$ WLS; Deming; vetoes) $\rightarrow$ metrology_pipeline.\{pdf,png\}.

Expected residual vs h $\rightarrow$ expected_residual_vs_height.\{pdf,png\}.

Expected T_2^{-1} vs $\Delta \ln \Phi$ $\rightarrow$ expected_T2_vs_height.\{pdf,png\}.

Residual QQ $\rightarrow$ residual_qqplot.\{pdf,png\}.

Per-lab forest $\rightarrow$ forest_vertical_clocks.\{pdf,png\}.

Intercept stability $\rightarrow$ intercept_stability.\{pdf,png\}.

Network map $\rightarrow$ map_dPhi_over_Phi_bounds.\{pdf,png\}.

Notebook	Inputs	Outputs
01_gr_subtraction_slope_fit.ipynb	<code>primary_schema.csv</code> $\rightarrow X_0 = g\Delta h/c^2$, $Y_0 = \delta y_{\text{res}}$	<code>expected_residual_vs_height.png</code> , WLS/HAC slope table, forest plot
02_clock_residual_simulator.ipynb	<code>noise_palette.yaml</code> , ε_Φ , N , Δh_{rms}	<code>sim_runs.csv</code> , ROC/power curves (Sec. 12)
03_common_mode_bayes.ipynb	UTC/TAI series, site logs	Coherence plots, $\ln B_{10}$ table, regional $ \delta\Phi/\Phi $ bounds (Sec. 12)
04_qubit_T2_gradient_fit.ipynb	T_2, T_1, T_ϕ datasets, exposure $X_\Phi = \Delta \ln \Phi$	T_2^{-1} vs X figure, $\hat{\beta} \pm 1\sigma$ table (Sec. 12)
05_cosmo_Ephi_fit.ipynb	$H(z)$, BAO/AP, $D_{\Delta t}H_0/c$, SN (shape-only)	Posterior $E_\Phi(z)$, $\Delta\chi^2$, 95% $ \varepsilon_\Phi^{\text{cos}} $ bounds (Sec. 12)
06_spectroscopy_drift_bounds.ipynb	Multi-year ν_a/ν_S , optical ratios R_{ab}	$ \dot{\Phi}/\Phi $ bound, $\dot{R}_{ab}$ null checks (Sec. 12)

Table 2 Notebook deliverables (inputs $\rightarrow$ outputs).

CI hooks & ledger packaging (helpers).

- `00_ledger_packager.py`: compute SHA-256 for raw/clean bundles; emit `ledger_row.json`; update `runs_ledger.csv`.
- `analysis.yaml`: freeze prereg parameters (Andrews bandwidth L , bootstrap sizes, veto thresholds, band $[f_1, f_2]$; policies from Secs. 5, 8).
- Continuous integration: rebuild figs/tables from `data/processed/` on commit; verify invariance/ablations checks (Secs. 6, 12); fail CI if any gate flips to “veto”.

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